



528 - Clinical Outcomes among Patients on Injectable CAB/RPV With and Without Viremia at 8 US Clinics

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Background

- Accelerating Implementation of Multilevel Strategies to Advance Long-Acting Injectables for Underserved Populations (**ALAI UP Project**) is a **HRSA-funded service delivery project** designed to help introduce and scale LAI ART at 8 clinics across the US.
- ALAI UP clinics varied** with respect to implementation stage, organizational type (HIV Service Organizations, academic medical centers, safety net hospitals, community-based organizations), and geographic region across US (South, SE, NE, Midwest, Mid-Atlantic).
- ALAI UP focuses on the delivery of **cabotegravir and rilpivirine (iCAB/RPV)**, administered as monthly or bimonthly intramuscular injections, currently the only complete LAI ART regimen.
- There is **limited data on iCAB/RPV use** from clinics caring for underserved populations outside of clinical trials and academic medical centers and in patients initiating iCAB/RPV with viremia.

Methods

Data Collection:

- Data extracted directly from electronic medical records and by chart review (as needed) for all adults with HIV who had ever received an iCAB/RPV injection at an ALAI UP clinic from 3/1/23 to 2/28/25
- Clinical events like test results or injection dates could occur earlier than this window.

Statistical Methods:

- The Kaplan-Meier survival estimator was used for analysis of time to first viral load (VL) <200 copies/mL, time to first VL >200 copies/mL, and time to iCAB/RPV discontinuation.
- The cumulative incidence function was used for competing risks for time to first failure and loss to follow-up, since a patient LTFU on iCAB/RPV may subsequently experience treatment failure which would otherwise show up as censoring if competing risks were not considered.

Study population across 8 clinics, stratified by VL prior to initiating iCAB/RPV shown below (Table 1)

Table 1. People with HIV at ALAI UP Clinics Ever On iCAB/RPV

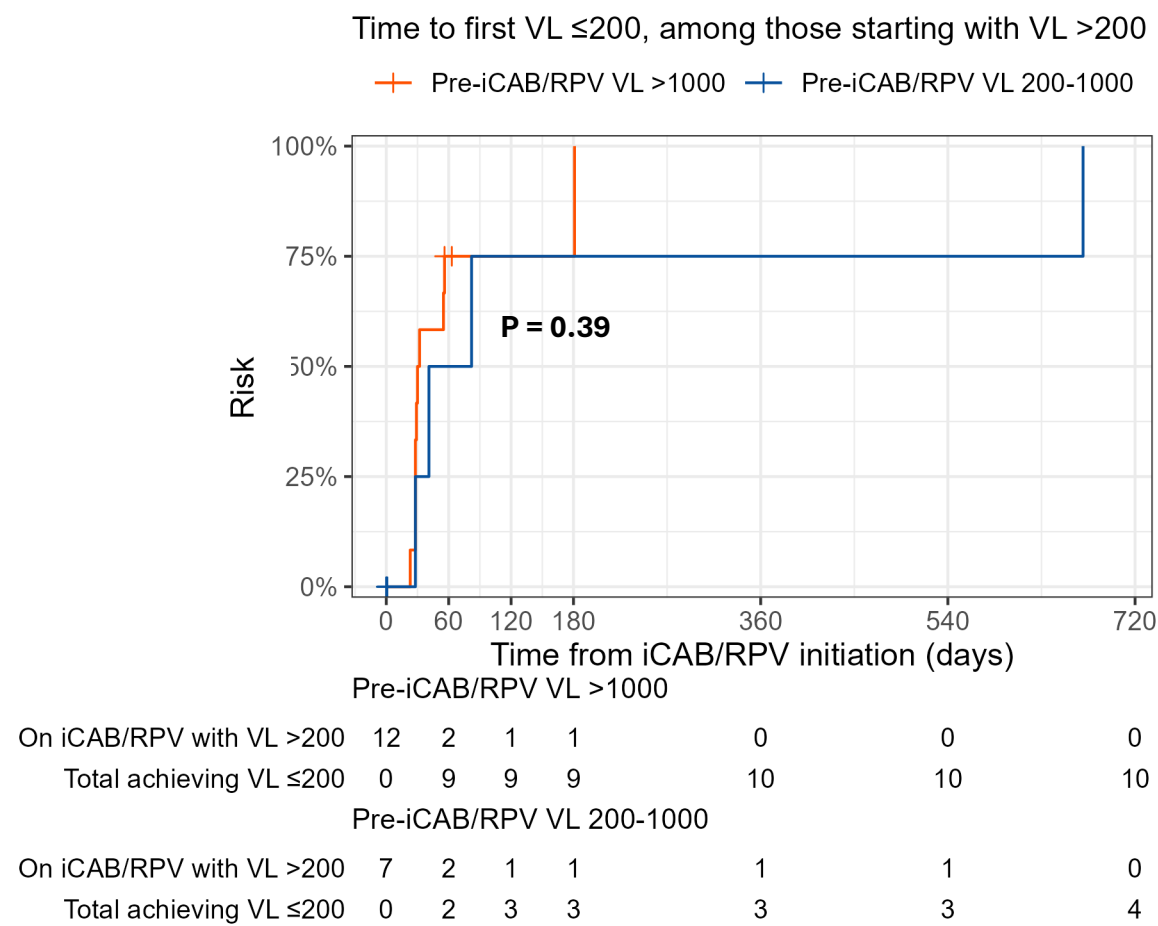
		Pre-iCAB/RPV VL					
Characteristic	Overall N = 344	<50 N = 302	50-200 N = 20	>200 N = 19	Missing N = 3		p-value
Age, years							
18-24	9 (2.6%)	8 (2.6%)	0 (0%)	1 (5.3%)	0 (0%)	>0.9	
25-34	85 (25%)	74 (25%)	5 (25%)	5 (26%)	1 (33%)		
35-44	126 (37%)	107 (35%)	11 (55%)	7 (37%)	1 (33%)		
45-54	71 (21%)	64 (21%)	2 (10%)	4 (21%)	1 (33%)		
55-64	39 (11%)	36 (12%)	1 (5.0%)	2 (11%)	0 (0%)		
65+	14 (4.1%)	13 (4.3%)	1 (5.0%)	0 (0%)	0 (0%)		
Race							
American Indian or Alaskan Native	1 (0.3%)	0 (0%)	0 (0%)	1 (5.3%)	0 (0%)	0.02	
Asian	1 (0.3%)	1 (0.3%)	0 (0%)	0 (0%)	0 (0%)		
Black	228 (66%)	204 (68%)	11 (55%)	10 (53%)	3 (100%)		
Native Hawaiian or Pacific Islander	0 (0%)	0 (0%)	0 (0%)	0 (0%)	0 (0%)		
White	60 (17%)	55 (18%)	2 (10%)	3 (16%)	0 (0%)		
Other	12 (3.5%)	7 (2.3%)	4 (20%)	1 (5.3%)	0 (0%)		
Unknown	42 (12%)	35 (12%)	3 (15%)	4 (21%)	0 (0%)		
Ethnicity							
Hispanic	79 (23%)	66 (22%)	7 (35%)	6 (32%)	0 (0%)	0.4	
Non-Hispanic	264 (77%)	235 (78%)	13 (65%)	13 (68%)	3 (100%)		
Unknown	1 (0.3%)	1 (0.3%)	0 (0%)	0 (0%)	0 (0%)		
Sex							
Male	279 (81%)	247 (82%)	13 (65%)	17 (89%)	2 (67%)	0.2	
Female	65 (19%)	55 (18%)	7 (35%)	2 (11%)	1 (33%)		

Research Questions, Results, and Context

Q1: How long does it take to achieve viral suppression, if initiating with viremia?

Definition: Viremia, Viral Suppression

- Pre-iCAB/RPV VL reported by clinic or most recent VL (within 3 months before to 2 weeks after first injection)
- Viremia defined as pre-iCAB/RPV VL >200 copies/mL
- Viral suppression defined as VL≤200 copies/mL



Key Findings

- People initiating with pre-iCAB/RPV VL >200 copies/mL: Quickly reached viral suppression
 - 79% by 12 weeks
 - 79% by 180 days
 - 90% by 360 days
- Subsequent rebound (VL >200) was rare (<3%) across pre-iCAB/RPV VL groups

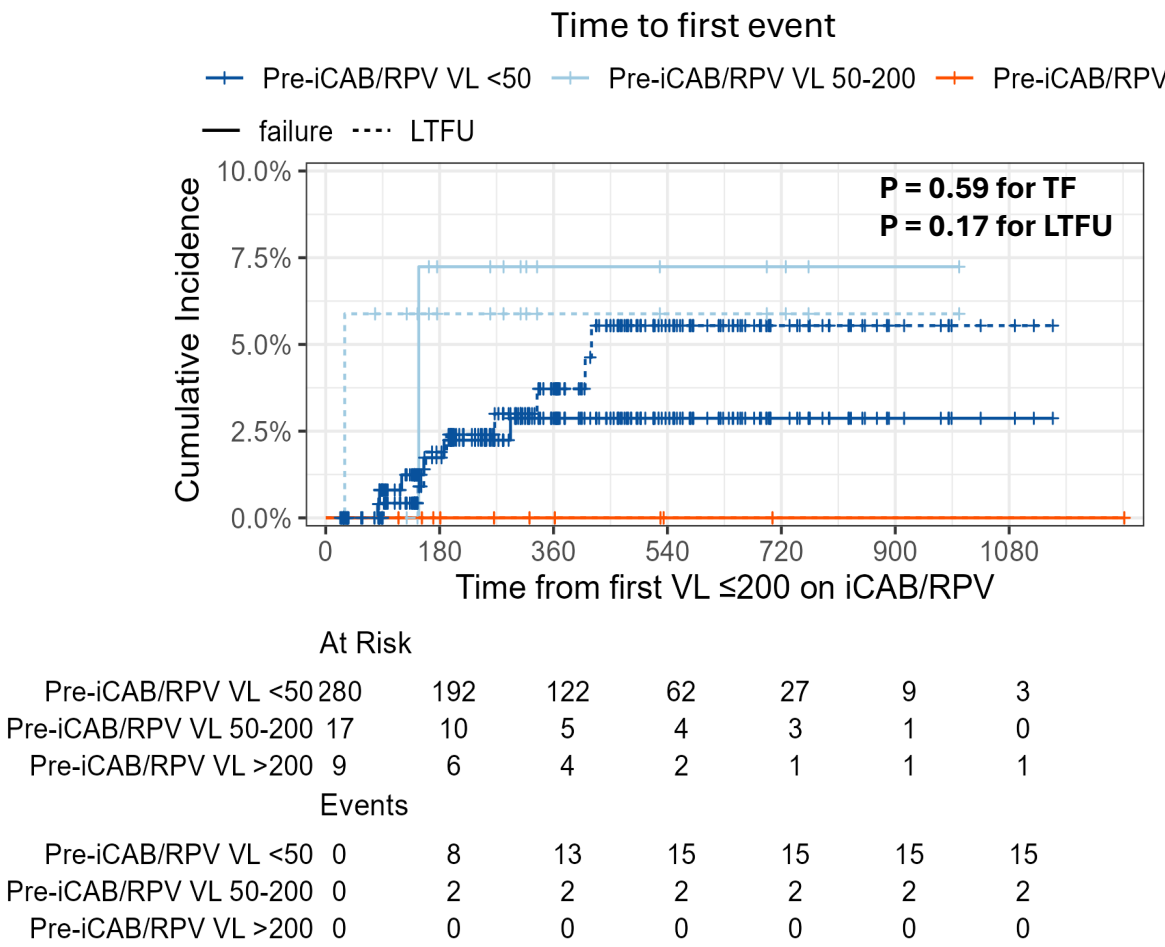
In Context

- 2 other studies: 80%-100% achieve viral suppression

Q2: How common is treatment failure while on iCAB/RPV?

Definition: Treatment Failure

- 1 VL >1000 copies/mL
- 2 consecutive VL >200 copies/mL
- Emergent drug resistance



Key Findings

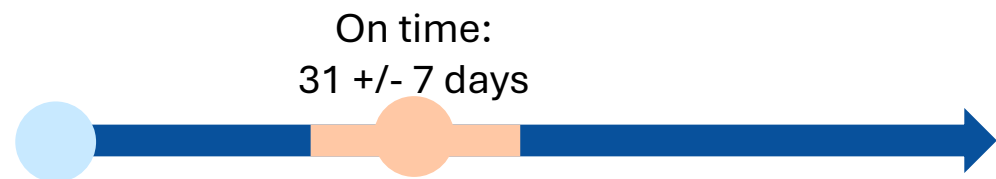
- People initiating with pre-iCAB/RPV VL >200 copies/mL: No treatment failures at 360 days
- People with pre-iCAB/RPV VL <50 copies/mL: 2.5% risk of treatment failure at 360 days (95% CI: 1.3%-6.4%)
- People with pre-iCAB/RPV VL 50-200 copies/mL: 7.2% risk of treatment failure at 360 days (95% CI 1.1%-47.8%); limitation: only 1 event
- 3 of the 7 people with treatment failure had emergent drug resistance

In Context:

- 13 other studies: 94-100% maintain viral suppression

Q3: How common are late injections?

Definition: On time second injections and monthly injections



Definition: On time bi-monthly injections



*Injection was not late if oral bridging used

On time status	Second injection N = 290	Monthly injections N = 217	Bimonthly injections N = 1,595	p-value
Early	9 (3.1%)	4 (1.8%)	215 (13%)	
On time	276 (95%)	207 (95%)	1,342 (84%)	<0.001
Late	5 (1.7%)	6 (2.8%)	38 (2.4%)	

Key Findings

- Regardless of injection type, most injections (98%) were early or on-time
- 1.2% (4/344) people had injections >30 days late, which required reloading

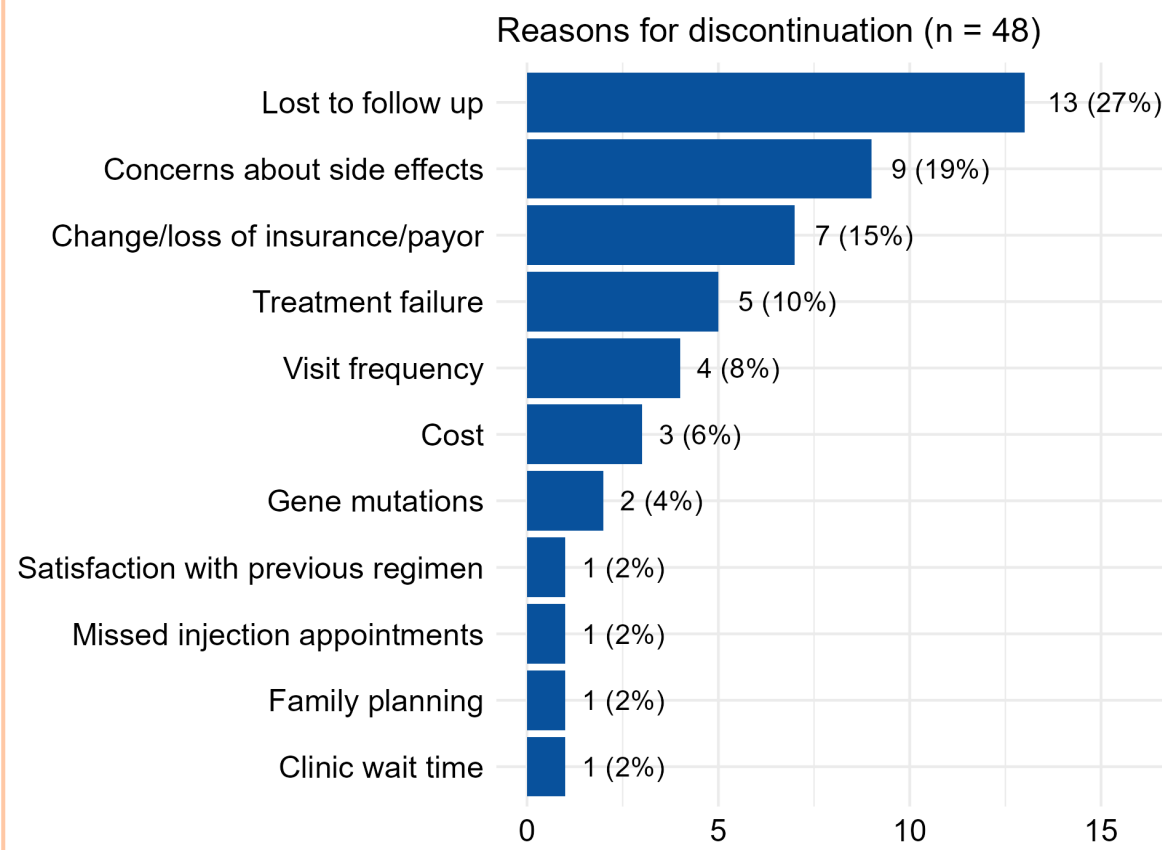
In Context:

- 9 other studies: 85-98% of injections on time

Q4: How common are discontinuations? Why do people discontinue iCAB/RPV?

Definition: Discontinuation

- Clinic reports person has stopped taking iCAB/RPV
- No iCAB/RPV injections 90 days after last injection



Key Findings

- 47/344 people (14%) discontinued iCAB/RPV by the end of the study period
- An additional 20 people transferred to other clinics but were sustained on iCAB/RPV after transfer
- Time on iCAB/RPV before discontinuation did not vary significantly by pre-iCAB/RPV VL or history of late injections (data not shown)

In Context:

- 10 other studies: 6-27% people discontinued

Clinics caring for underserved populations with health-related social needs can successfully deliver iCAB/RPV to patients with and without viremia and achieve clinical outcomes comparable to clinical trials and demonstration projects at academic medical centers.

Conclusions

- Clinical outcomes at ALAI UP clinics through 2 years were similar to outcomes from clinical trials and academic medical centers.
- Most people on iCAB/RPV had sustained viral suppression while receiving injections, and most injections were received on time.
- Discontinuations are not uncommon even in the setting of effective viral suppression and adherence.
- Limitation: Limited power to detect significant differences by different sub-groups due to few people who initiated iCAB/RPV with viremia and experienced elevated VL outcome.

Acknowledgements

The Health Resources and Services Administration (HRSA), Department of Health and Human Services (HHS) provided financial support for this project. The award provided 100% of total costs and totaled \$7,450,000. The contents are those of the presenters. They may not reflect the policies of HRSA, HHS, or the U.S. Government. We are grateful for the collaboration and commitment of our partner organizations, including Columbia University Irving Medical Center, New York City Health Department, Hunter Alliance for Research and Translation at Hunter College of the City University of New York, Albert Einstein College of Medicine, and the Southeast AIDS Education and Training Center. We also thank our demonstration site partners for their essential contributions to this work: Abounding Prosperity; Baltimore City Health Department; Coastal Family Health Center; Florida Department of Health, Sunshine Care Center; Jack Martin Fund Center, Mount Sinai Institute for Advanced Medicine; Positive Impact Health Centers; San Antonio AIDS Foundation; and Sinai Infectious Disease Center. Collectively, these collaborators comprise what we refer to as the “ALAI UP Project Team” in the author list.